

JEE Mains 2019 Chapter wise Question Bank

Functions – Questions

Q1

For $x \in \mathbf{R} - \{0, 1\}$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x)$

$= \frac{1}{1-x}$ be three given functions. If a function, $J(x)$

satisfies $(f_2 \circ f \circ f_1)(x) = f_3(x)$ then $J(x)$ is equal to:

- (1) $f_3(x)$ (2) $\frac{1}{x} f_3(x)$
 (3) $f_2(x)$ (4) $f_1(x)$

9 Jan Morning

Q2

If the fractional part of the number $\frac{2^{403}}{15}$ is $\frac{k}{15}$, then k is equal to:

- (1) 6 (2) 8
 (3) 4 (4) 14

9 Jan Morning

Q3

Let $A = \{x \in \mathbf{R} : x \text{ is not a positive integer}\}$. Define a

function $f: A \rightarrow \mathbf{R}$ as $f(x) = \frac{2x}{x-1}$, then f is:

- (1) not injective
 (2) neither injective nor surjective
 (3) surjective but not injective
 (4) injective but not surjective

9 Jan Evening

Q4

Let $\mathbf{N}$ be the set of natural numbers and two functions f and g be defined as $f, g: \mathbf{N} \rightarrow \mathbf{N}$ such that

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$$

and $g(n) = n - (-1)^n$. Then $f \circ g$ is:

- (1) onto but not one-one.
 (2) one-one but not onto.
 (3) both one-one and onto.
 (4) neither one-one nor onto.

10 Jan Evening

Q5

Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = \frac{x}{1+x^2}$, $x \in \mathbf{R}$. Then

the range of f is:

- (1) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (2) $\mathbf{R} - [-1, 1]$
 (3) $\mathbf{R} - \left[-\frac{1}{2}, \frac{1}{2}\right]$ (4) $(-1, 1) - \{0\}$

11 Jan Morning

Q6

Let a function $f: (0, \infty) \rightarrow (0, \infty)$ be defined by

$$f(x) = \left| 1 - \frac{1}{x} \right|. \text{ Then } f \text{ is:}$$

- (1) not injective but it is surjective
 (2) injective only
 (3) neither injective nor surjective
 (4) both injective as well as surjective

11 Jan Evening

Functions

Q7

If $f(x) = \log_e \left(\frac{1-x}{1+x} \right)$, $|x| < 1$, then $f\left(\frac{2x}{1+x^2}\right)$ is equal to :

- (1) $2f(x)$ (2) $2f(x^2)$
(3) $(f(x))^2$ (4) $-2f(x)$

8 April Morning

Q8

Let $f(x) = a^x$ ($a > 0$) be written as $f(x) = f_1(x) + f_2(x)$, where $f_1(x)$ is an even function and $f_2(x)$ is an odd function. Then $f_1(x+y) + f_1(x-y)$ equals :

- (1) $2f_1(x)f_1(y)$ (2) $2f_1(x+y)f_1(x-y)$
(3) $2f_1(x)f_2(y)$ (4) $2f_1(x+y)f_2(x-y)$

8 April Evening

Q9

If the function $f: \mathbb{R} - \{1, -1\} \rightarrow A$ defined by $f(x) = \frac{x^2}{1-x^2}$, is

surjective, then A is equal to:

- (1) $\mathbb{R} - \{-1\}$ (2) $[0, \infty)$
(3) $\mathbb{R} - [-1, 0)$ (4) $\mathbb{R} - (-1, 0)$

9 April Morning

Q10

Let $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$, where the function f satisfies

$f(x+y) = f(x)f(y)$ for all natural numbers x, y and $f(1) = 2$.

Then the natural number 'a' is:

- (1) 2 (2) 16 (3) 4 (4) 3

9 April Morning

Q11

The domain of the definition of the function

$f(x) = \frac{1}{4-x^2} + \log_{10}(x^3 - x)$ is:

- (1) $(-1, 0) \cup (1, 2) \cup (3, \infty)$
(2) $(-2, -1) \cup (-1, 0) \cup (2, \infty)$
(3) $(-1, 0) \cup (1, 2) \cup (2, \infty)$
(4) $(1, 2) \cup (2, \infty)$

9 April Evening

Q12

JEE Mains 2019 Chapter wise Question Bank

Let $f(x) = x^2, x \in \mathbb{R}$. For any $A \subseteq \mathbb{R}$, define $g(A) = \{x \in \mathbb{R} : f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true ?

- (1) $g(f(S)) \neq S$ (2) $f(g(S)) = S$
(3) $g(f(S)) = g(S)$ (d) $f(g(S)) \neq f(S)$

10 April Morning

Q13

Let $f(x) = \log_e(\sin x)$, ($0 < x < \pi$) and $g(x) = \sin^{-1}(e^{-x})$, ($x \geq 0$). If α is a positive real number such that $a = (fog)'(\alpha)$ and $b = (fog)(\alpha)$, then:

- (1) $a\alpha^2 + b\alpha + a = 0$ (2) $a\alpha^2 - b\alpha - a = 1$
(3) $a\alpha^2 - b\alpha - a = 0$ (4) $a\alpha^2 + b\alpha - a = -2a^2$

10 April Evening

Q14

For $x \in \left(0, \frac{3}{2}\right)$, let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$.

If $\phi(x) = ((hof)og)(x)$, then $\phi\left(\frac{\pi}{3}\right)$ is equal to :

- (1) $\tan \frac{\pi}{12}$ (2) $\tan \frac{11\pi}{12}$ (3) $\tan \frac{7\pi}{12}$ (4) $\tan \frac{5\pi}{12}$

12 April Morning

JEE Mains 2019 Chapter wise Question Bank

Functions - Solutions

Q1

(1) The given relation is

$$(f_2 \circ J \circ f_1)(x) = f_3(x) = \frac{1}{1-x}$$

$$\Rightarrow (f_2 \circ J)(f_1(x)) = \frac{1}{1-x}$$

$$\Rightarrow (f_2 \circ J)\left(\frac{1}{x}\right) = \frac{1}{1 - \frac{1}{x}} = \frac{\frac{1}{x}}{\frac{x-1}{x}} \quad \left[\because f_1(x) = \frac{1}{x} \right]$$

$$\Rightarrow (f_2 \circ J)(x) = \frac{x}{x-1} \quad \left[\frac{1}{x} \text{ is replaced by } x \right]$$

$$\Rightarrow f_2(J(x)) = \frac{x}{x-1}$$

$$\Rightarrow 1 - J(x) = \frac{x}{x-1} \quad [\because f_2(x) = 1-x]$$

$$\therefore J(x) = 1 - \frac{x}{x-1} = \frac{1}{1-x} = f_3(x)$$

9 Jan Morning

Q2

$$\begin{aligned} (2) \quad 2^{403} &= 2^{400} \cdot 2^3 \\ &= 2^{4 \times 100} \cdot 2^3 \\ &= (2^4)^{100} \cdot 8 \\ &= 8(2^4)^{100} = 8(16)^{100} \\ &= 8(1+15)^{100} \\ &= 8 + 15\mu \end{aligned}$$

When 2^{403} is divided by 15, then remainder is 8.Hence, fractional part of the number is $\frac{8}{15}$ Therefore value of k is 8

9 Jan Morning

Q3

(4) As $A = \{x \in R : x \text{ is not a positive integer}\}$ A function $f: A \rightarrow R$ given by $f(x) = \frac{2x}{x-1}$

$$f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$$

So, f is one-one.As $f(x) \neq 2$ for any $x \in A \Rightarrow f$ is not onto.Hence f is injective but not surjective.

9 Jan Evening

Q4

Functions

$$(1) f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$

$$g(n) = \begin{cases} 2, & n=1 \\ 1, & n=2 \\ 4, & n=3 \\ 3, & n=4 \\ 6, & n=5 \\ 5, & n=6 \end{cases}$$

Then,
$$f(g(n)) = \begin{cases} \frac{g(n)+1}{2}, & \text{if } g(n) \text{ is odd} \\ \frac{g(n)}{2}, & \text{if } g(n) \text{ is even} \end{cases}$$

$$f(g(n)) = \begin{cases} 1, & n=1 \\ 1, & n=2 \\ 2, & n=3 \\ 2, & n=4 \\ 3, & n=5 \\ 3, & n=6 \\ \vdots & \vdots \\ \vdots & \vdots \\ \vdots & \vdots \end{cases}$$

$\Rightarrow fog$ is onto but not one - one

10 Jan Evening

Q5

JEE Mains 2019 Chapter wise Question Bank

$$(1) f(x) = \frac{x}{1+x^2}, x \in R$$

$$\text{Let, } y = \frac{x}{1+x^2}$$

$$\Rightarrow yx^2 - x + y = 0 \Rightarrow x = \frac{1 \pm \sqrt{1-4y^2}}{2}$$

$$\Rightarrow 1 - 4y^2 \geq 0$$

$$\Rightarrow 1 \geq 4y^2$$

$$\Rightarrow |y| \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq y \leq \frac{1}{2}$$

$$\Rightarrow \text{The range of } f \text{ is } \left[-\frac{1}{2}, \frac{1}{2}\right].$$

11 Jan Morning

Q6

$$(\text{Bonus}) f: (0, \infty) \rightarrow (0, \infty)$$

$$f(x) = \left| 1 - \frac{1}{x} \right| \text{ is not a function}$$

$$\because f(1) = 0 \text{ and } 1 \in \text{domain but } 0 \notin \text{co-domain}$$

Hence, $f(x)$ is not a function.

11 Jan Evening

Q7

Functions

$$(1) f(x) = \log\left(\frac{1-x}{1+x}\right), |x| < 1$$

$$f\left(\frac{2x}{1+x^2}\right) = \log\left(\frac{1 - \frac{2x}{1+x^2}}{1 + \frac{2x}{1+x^2}}\right)$$

$$= \log\left(\frac{1+x^2-2x}{1+x^2+2x}\right)$$

$$= \log\left(\frac{1-x}{1+x}\right)^2$$

$$= 2 \log\left(\frac{1-x}{1+x}\right)$$

$$= 2f(x)$$

8 April Morning

Q8

(1) Given function can be written as

$$f(x) = a^x = \left(\frac{a^x + a^{-x}}{2}\right) + \left(\frac{a^x - a^{-x}}{2}\right)$$

where $f_1(x) = \frac{a^x + a^{-x}}{2}$ is even function

$f_2(x) = \frac{a^x - a^{-x}}{2}$ is odd function

$$\Rightarrow f_1(x+y) + f_1(x-y)$$

$$= \left(\frac{a^{x+y} + a^{-x-y}}{2}\right) + \left(\frac{a^{x-y} + a^{-x+y}}{2}\right)$$

$$= \frac{1}{2} [a^x(a^y + a^{-y}) + a^{-x}(a^y + a^{-y})]$$

$$= \frac{(a^x + a^{-x})(a^y + a^{-y})}{2} = 2f_1(x) \cdot f_1(y)$$

8 April Evening

Q9

JEE Mains 2019 Chapter wise Question Bank

$$(3) f(x) = \frac{x^2}{1-x^2}$$

$$\Rightarrow f(-x) = \frac{x^2}{1-x^2} = f(x)$$

$$f'(-x) = \frac{2x}{(1-x^2)^2}$$

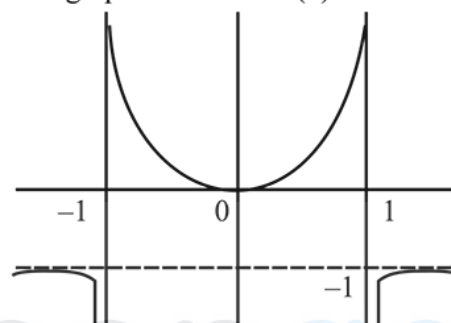
$\therefore f(x)$ increases in $x \in (0, \infty)$

Also $f(0) = 0$ and

$\lim_{x \rightarrow \pm\infty} f(x) = -1$ and $f(x)$ is even function

$\therefore$ Set $A = \mathbb{R} - [-1, 0)$

And the graph of function $f(x)$ is



9 April Morning

Q10

$$(4) \because f(x+y) = f(x) \cdot f(y)$$

$$\Rightarrow \text{Let } f(x) = t^x$$

$$\because f(1) = 2$$

$$\therefore t = 2$$

$$\Rightarrow f(x) = 2^x$$

$$\text{Since, } \sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$$

$$\text{Then, } \sum_{k=1}^{10} 2^{a+k} = 16(2^{10} - 1)$$

$$\Rightarrow 2^a \sum_{k=1}^{10} 2^k = 16(2^{10} - 1)$$

$$\Rightarrow 2^a \times \frac{((2^{10}) - 1) \times 2}{(2 - 1)} = 16 \times (2^{10} - 1) \Rightarrow 2 \cdot 2^a = 16$$

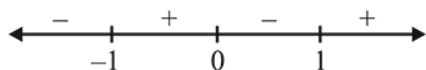
$$\Rightarrow a = 3$$

9 April Morning

Functions

Q11

- (3) To determine domain, denominator $\neq 0$ and $x^3 - x > 0$
 i.e., $4 - x^2 \neq 0$ $x \neq \pm 2$... (1)
 and $x(x-1)(x+1) > 0$



$$x \in (-1, 0) \cup (1, \infty) \quad \dots (2)$$

Hence domain is intersection of (1) & (2).

$$\text{i.e., } x \in (-1, 0) \cup (1, 2) \cup (2, \infty)$$

9 April Evening

Q12

- (3) $f(x) = x^2$; $x \in \mathbb{R}$
 $g(A) = \{x \in \mathbb{R} : f(x) \in A\}$ $S = [0, 4]$
 $g(S) = \{x \in \mathbb{R} : f(x) \in S\}$
 $= \{x \in \mathbb{R} : 0 \leq x^2 \leq 4\} = \{x \in \mathbb{R} : -2 \leq x \leq 2\}$
 $\therefore g(S) \neq S \therefore f(g(S)) \neq f(S)$
 $g(f(S)) = \{x \in \mathbb{R} : f(x) \in f(S)\}$
 $= \{x \in \mathbb{R} : x^2 \in S^2\} = \{x \in \mathbb{R} : 0 \leq x^2 \leq 16\}$
 $= \{x \in \mathbb{R} : -4 \leq x \leq 4\}$
 $\therefore g(f(S)) \neq g(S)$
 $\therefore g(f(S)) = g(S)$ is incorrect.

10 April Morning

Q13

- (2) $f(x) = \ln(\sin x)$, $g(x) = \sin^{-1}(e^{-x})$
 $\Rightarrow f(g(x)) = \ln(\sin(\sin^{-1} e^{-x})) = -x$
 $\Rightarrow f(g(x)) = -\alpha$
 But given that $(f \circ g)(\alpha) = b$
 $\therefore -\alpha = b$ and $f'(g(\alpha)) = a$, i.e., $a = -1$
 $\therefore a\alpha^2 - b\alpha - a = -\alpha^2 + \alpha^2 - (-1)$
 $\Rightarrow a\alpha^2 - b\alpha - a = 1.$

10 April Evening

Q14

JEE Mains 2019 Chapter wise Question Bank

- (2) $\because \phi(x) = ((hof)og)(x)$
 $\therefore \phi\left(\frac{\pi}{3}\right) = h\left(f\left(g\left(\frac{\pi}{3}\right)\right)\right) = h(f(\sqrt{3})) = h(3^{1/4})$
 $= \frac{1-\sqrt{3}}{1+\sqrt{3}} = -\frac{1}{2}(1+3-2\sqrt{3}) = \sqrt{3}-2 = -(-\sqrt{3}+2)$
 $= -\tan 15^\circ = \tan(180^\circ - 15^\circ) = \tan\left(\pi - \frac{\pi}{12}\right) = \tan \frac{11\pi}{12}$

12 April Morning